WORKED SOLUTION

STEP III 2021, Question 12

Probability and Statistics • Variance of Sums

STEP III 2021, Question 12

Read the question before you go on. It is in the STEP III paper in OCR's download of the 2021 STEP papers (a zip file):

www.ocr.org.uk/Images/696588-2021-past-papers.zip

This question has a full guided treatment in *Beyond STEP*, Section 12.4, *Independent in Pairs, Not All Together*: https://beyondstep.keithmcnulty.org/ch12/pairwise-independence.html

The key idea

Write the team's score as a sum of indicator-type variables, one for each pair of players, and expand the variance of the sum. Everything then rests on how two *different* pairs interact. If the pairs have no player in common they are plainly independent, so the real work is with pairs that share one player. In (i) such pairs turn out to be independent, so the cross terms vanish. In (iii) they are not, and the identity in (ii) is the tool for counting what they contribute.

Solution

Part (i)

Each die is fair and the players roll independently. Two given players roll the same number with probability $\frac{1}{6}$, whatever the first of them rolls. So

$$\mathrm{P}(X_{12} = 1) = \mathrm{P}(X_{23} = 1) = \tfrac{1}{6}.$$

Both $X_{12} = 1$ and $X_{23} = 1$ happen exactly when players 1, 2 and 3 all roll the same number, so

$$\mathrm{P}(X_{12} = 1,\ X_{23} = 1) = \tfrac{1}{6} \cdot \tfrac{1}{6} = \tfrac{1}{36} = \mathrm{P}(X_{12} = 1)\,\mathrm{P}(X_{23} = 1).$$

Subtracting this from the marginal probabilities gives the other three cells. For example, $\mathrm{P}(X_{12} = 1, X_{23} = 0) = \frac{1}{6} - \frac{1}{36} = \frac{1}{6} \cdot \frac{5}{6}$, and similarly $\mathrm{P}(X_{12} = 0, X_{23} = 1) = \frac{5}{6} \cdot \frac{1}{6}$ and $\mathrm{P}(X_{12} = 0, X_{23} = 0) = 1 - \frac{1}{6} - \frac{1}{6} + \frac{1}{36} = \frac{5}{6} \cdot \frac{5}{6}$. So the joint distribution is the product of the marginals, and X_{12} and X_{23} are independent.

Ask yourself Why does knowing $X_{12} = 1$ tell us nothing about X_{23}?

Given that players 1 and 2 match, player 2's number is still uniform on $\{1, \ldots, 6\}$, and player 3 matches it with probability $\frac{1}{6}$, exactly as before. The shared player carries no information because every face is equally likely.

Let $T = \sum_{i<j} X_{ij}$ be the total score, a sum of $N = \binom{n}{2}$ terms. Each X_{ij} is Bernoulli with parameter $\frac{1}{6}$, so $\mathrm{E}(X_{ij}) = \frac{1}{6}$ and $\mathrm{Var}(X_{ij}) = \frac{1}{6} \cdot \frac{5}{6} = \frac{5}{36}$. Hence

$$\mathrm{E}(T) = \frac{N}{6} = \frac{n(n-1)}{12}.$$

For the variance, take any two different pairs. If they have no player in common, they depend on disjoint sets of dice and so are independent. If they share one player, they are independent by the argument above, after relabelling the players. So the X_{ij} are pairwise independent, every covariance is zero, and the variance of the sum is the sum of the variances:

$$\mathrm{Var}(T) = N \cdot \frac{5}{36} = \frac{5n(n-1)}{72}.$$

Part (ii)

Let $S = Y_1 + \cdots + Y_m$. Then $\mathrm{E}(S) = 0$ by linearity, so $\mathrm{Var}(S) = \mathrm{E}(S^2) - 0^2 = \mathrm{E}(S^2)$. Expanding the square, each Y_i^2 appears once and each product $Y_i Y_j$ with $i < j$ appears twice:

$$S^2 = \sum_{i=1}^{m} Y_i^2 + 2 \sum_{i=1}^{m-1} \sum_{j=i+1}^{m} Y_i Y_j.$$

Taking expectations and using linearity gives the required formula.

Part (iii)

For two players, $\mathrm{P}(\text{same even number}) = \frac{3}{36} = \frac{1}{12}$, and likewise $\mathrm{P}(\text{same odd number}) = \frac{1}{12}$. So

$$\mathrm{P}(Z_{ij} = 1) = \mathrm{P}(Z_{ij} = -1) = \tfrac{1}{12}, \qquad \mathrm{P}(Z_{ij} = 0) = \tfrac{5}{6}.$$

Not independent. If $Z_{12} = 1$ and $Z_{23} = -1$, player 2's number would have to be both even and odd. So

$$\mathrm{P}(Z_{12} = 1,\ Z_{23} = -1) = 0 \neq \tfrac{1}{12} \cdot \tfrac{1}{12} = \mathrm{P}(Z_{12} = 1)\,\mathrm{P}(Z_{23} = -1),$$

and Z_{12} and Z_{23} are not independent.

Mean. By symmetry $\mathrm{E}(Z_{ij}) = \frac{1}{12} - \frac{1}{12} = 0$, so the total score $U = \sum_{i<j} Z_{ij}$ has mean 0.

Variance. Every Z_{ij} has mean zero, so we can use (ii) with the $N = \binom{n}{2}$ variables Z_{ij}. We need two kinds of expectation.

- Squares: Z_{ij}^2 is 1 exactly when players i and j match, so $\mathrm{E}(Z_{ij}^2) = \frac{1}{6}$.

- Products of two different pairs. If the pairs have no player in common, the two variables are independent with mean zero, so $\mathrm{E}(Z_{ij}Z_{kl}) = \mathrm{E}(Z_{ij})\,\mathrm{E}(Z_{kl}) = 0$. If they share a player, say $Z_{12}Z_{23}$, the product is nonzero only when all three players roll the same number. Then both factors are $+1$ (even) or both are -1 (odd), so the product is 1. Hence $\mathrm{E}(Z_{12}Z_{23}) = \frac{1}{36}$.

? **Ask yourself** How many unordered pairs of pairs share exactly one player?

Choose the shared player in n ways, then choose its two partners from the remaining $n-1$ players in $\binom{n-1}{2}$ ways. This gives $n\binom{n-1}{2} = \frac{1}{2}n(n-1)(n-2)$ such pairs of pairs, each counted once, which is what the double sum in (ii) needs.

Putting this together,

$$\begin{aligned}\mathrm{Var}(U) &= \binom{n}{2}\cdot\frac{1}{6} + 2\cdot\frac{n(n-1)(n-2)}{2}\cdot\frac{1}{36}\\ &= \frac{n(n-1)}{36}\big(3+(n-2)\big)\\ &= \frac{n(n-1)(n+1)}{36} = \frac{1}{36}n(n^2-1),\end{aligned}$$

as required.

Checks

- For the example roll $3,3,3,3,6,6,2$, the pair count gives $6+1=7$ and the signed count gives $-6+1=-5$, matching the question.
- With $n=2$ there is only one pair, so $\mathrm{Var}(U) = \mathrm{E}(Z_{12}^2) = \frac{1}{6}$, and the formula gives $\frac{2\cdot 3}{36} = \frac{1}{6}$.
- For $n=3$, $U = \pm 3$ when all three dice agree (probability $\frac{1}{72}$ each). Otherwise exactly one pair matches and $U = \pm 1$, with total probability $\frac{15}{36}$. So $\mathrm{E}(U^2) = 9\cdot\frac{1}{36} + \frac{15}{36} = \frac{2}{3}$, agreeing with $\frac{3\cdot 8}{36}$. Exact enumeration by computer of all outcomes for $n \le 6$ confirms both means and both variances; for example $n=6$ gives $\frac{25}{12}$ in (i) and $\frac{35}{6}$ in (iii).

Common slips

- Claiming independence of X_{12} and X_{23} after checking only one cell, without saying why the other cells follow. For two-valued variables they do follow, but say so.
- In (iii), forgetting that pairs sharing a player contribute $+\frac{1}{36}$ whether the common number is even or odd. The minus signs cancel in the product.
- Miscounting the overlapping pairs of pairs, for example using $n(n-1)(n-2)$ (ordered) and then also multiplying by 2.

Beyond STEP

Part (i) is a classic example of variables that are *pairwise* independent but not *mutually* independent: if $X_{12} = X_{23} = 1$ then $X_{13} = 1$ is forced. Pairwise independence is all a variance calculation needs, an idea used in randomised algorithms and in the birthday problem. Part (iii) shows the general method of covariances of sums when the terms are dependent. Both are developed in Blitzstein and Hwang, *Introduction to Probability*.

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